Experiment No. 2

Title

File and Directory Management and File Permissions in Linux

Aim

To study and practice Linux commands for file and directory management, understand Linux file and directory permissions, and create and edit files using text editors such as `nano` and `vi`.

Theory

Linux File System

Linux follows a hierarchical file system starting from the root directory `/`. All files and directories are arranged in a tree structure, allowing efficient organization and secure access control.

Files and Directories

- **Files** store data such as text, programs, or scripts.
- **Directories** store files and subdirectories.

File and Directory Permissions

Linux uses permissions to control access:

- **Read (r)** – Read file contents / list directory
- **Write (w)** – Modify file or directory
- **Execute (x)** – Execute file or access directory

Permissions apply to:

- **Owner**
- **Group**
- **Others**

Text Editors

- **nano:** Simple and beginner-friendly text editor
- **vi:** Powerful editor with command and insert modes

Commands Used

Commands	Description
pwd	Display current directory
ls	List files and directories
ls -l	Display detailed file information
cd	Change directory
mkdir	Create directory
rmdir	Remove empty directory
touch	Create empty file
cat	Display file contents
nano	Edit files using nano editor
vi	Edit files using vi editor
chmod	Change file permissions
stat	Display file status

Procedure / Steps

Step 1: Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Display the Current Directory using pwd command.

Step 3: List Files and Directories using:

- ls
- ls -l

```
erin@erin-VirtualBox: ~
erin@erin-VirtualBox:~$ pwd
/home/erin
erin@erin-VirtualBox:~$ ls
Desktop  Documents  Downloads  erin  lab_work  Music  Pictures  Public  snap  Templates  Videos
erin@erin-VirtualBox:~$ ls -l
total 44
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Desktop
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Documents
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Downloads
drwxrwxr-x 2 erin erin 4096 Feb 10 11:57 erin
drwxrwxr-x 2 erin erin 4096 Feb  6 14:09 lab_work
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Music
drwxr-xr-x 3 erin erin 4096 Feb  6 14:04 Pictures
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Public
drwx------ 5 erin erin 4096 Feb 17 10:38 snap
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Templates
drwxr-xr-x 2 erin erin 4096 Feb  6 13:49 Videos
erin@erin-VirtualBox:~$
```

Step 4: Create a New Directory

mkdir linux_lab

cd linux_lab

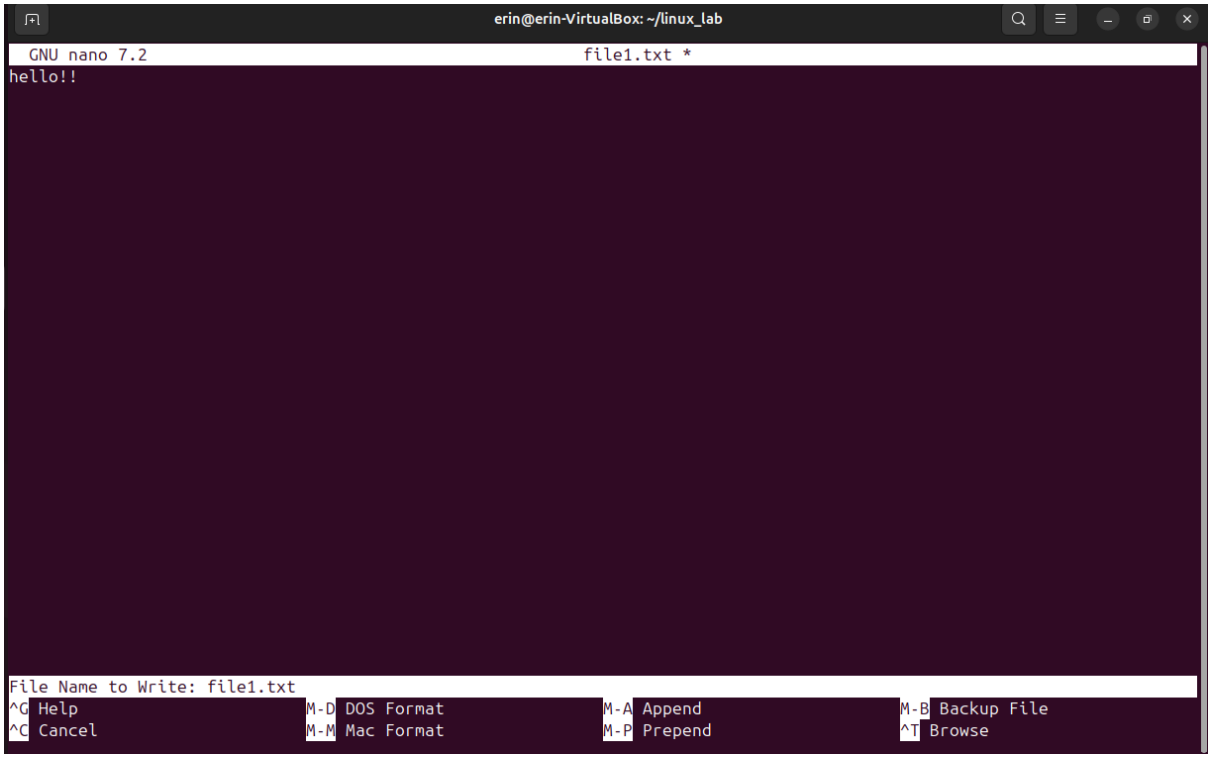
Step 5: Create Files using:

touch file1.txt file2.txt

```
erin@erin-VirtualBox:~$ mkdir linux_lab
erin@erin-VirtualBox:~$ cd linux_lab
erin@erin-VirtualBox:~/linux_lab$ touch file1.txt file2.txt
erin@erin-VirtualBox:~/linux_lab$
```

Step 6: Edit File Using: nano file1.txt

- Type some text
- Press **Ctrl + O** → Enter (save)
- Press **Ctrl + X** (exit)



Step 7: Display File Content Using:

- cat file1.txt

Step 8: Check File Permissions

- ls -l

Step 9: Change File Permissions using:

- chmod 755 file1.txt

Step 10: Verify Updated Permissions Using:

- ls -l file1.txt

  we will observe permission format: -rw-r--r—

```
erin@erin-VirtualBox:~/linux_lab$ ls -l
total 8
-rw-rw-r-- 1 erin erin 8 Mar  2 14:25 file1.txt
-rw-rw-r-- 1 erin erin 8 Mar  2 14:28 file1.txt.save
-rw-rw-r-- 1 erin erin 0 Mar  2 14:22 file2.txt
erin@erin-VirtualBox:~/linux_lab$ chmod 755 file1.txt
erin@erin-VirtualBox:~/linux_lab$ ls -l file1.txt
-rwxr-xr-x 1 erin erin 8 Mar  2 14:25 file1.txt
erin@erin-VirtualBox:~/linux_lab$
```

Outcomes

- ✓ Successful creation of directories and files
- ✓ Files edited using nano and vi
- ✓ File contents displayed using cat
- ✓ File permissions viewed and modified using chmod

Result

- ❖ Thus, file and directory management commands were successfully executed in Linux. The student learned to create, edit, and manage files and directories, understand file permissions, and use text editors such as nano and vi.